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| | | | |
|---|---|---|---|
| Patient Name: | | Patient DOB: | |
| Date Collected: | | Referring Physician: | |
| Biopsy Site: | **Axilla** | Primary Diagnosis: | **Melanoma** |
| Sample ID: | | MoCha ID: | |
| Telephone: | | Fax: | |
| Tumor Content (%): | **70%** | | |

Tumor content assessment performed by Van Andel Research Institute; Grand Rapids; MI 49503

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## Report Highlights

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## Relevant Melanoma Findings

| Gene | Finding |
|---|---|
| BRAF | ***BRAF V600E*** |
| KIT | None detected |
| NRAS | ***NRAS G13D*** |
| NTRK1 | None detected |
| NTRK2 | None detected |
| NTRK3 | None detected |

## Relevant Biomarkers

| Tier | Genomic Alteration | Relevant Therapies (In this cancer type) | Relevant Therapies (In other cancer type) | Clinical Trials |
|---|---|---|---|---|
| IA | *BRAF V600E* | **atezolizumab + cobimetinib + vemurafenib** [1]<br>**binimetinib + encorafenib** [1]<br>**cetuximab + encorafenib** [1]<br>**cobimetinib + vemurafenib** [1]<br>**dabrafenib** [1]<br>**dabrafenib + trametinib** [1]<br>**trametinib** [1]<br>**vemurafenib** [1]<br>dabrafenib + pembrolizumab + trametinib<br>encorafenib | **binimetinib + encorafenib** [1]<br>**cetuximab + encorafenib** [1]<br>**dabrafenib** [1]<br>**dabrafenib + trametinib** [1]<br>**trametinib** [1]<br>cobimetinib + vemurafenib<br>encorafenib + panitumumab<br>selumetinib<br>vemurafenib | 48 |

**Public data sources included in relevant therapies: FDA**1, NCCN

**Tier Reference:** Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

## Relevant Biomarkers (continued)

| Tier | Genomic Alteration | Relevant Therapies (In this cancer type) | Relevant Therapies (In other cancer type) | Clinical Trials |
|---|---|---|---|---|
| IA | *NRAS G13D* | binimetinib | None | 22 |

**Public data sources included in relevant therapies: FDA**1, NCCN
**Tier Reference:** Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

## Variant Details

### DNA Sequence Variants

| Gene | Amino Acid Change | Coding | Variant ID | Locus | Allele Frequency | Transcript | Variant Effect |
|---|---|---|---|---|---|---|---|
| NRAS | p.(G13D) | c.38G>A | COSM573 | chr1:115258744 | 10.48% | NM_002524.4 | missense |
| BRAF | p.(V600E) | c.1799T>A | COSM476 | chr7:140453136 | 47.82% | NM_004333.4 | missense |

## Comments:

N/A

Signed By: ___________________________________________________________________

## Biomarker Descriptions

**BRAF (B-Raf proto-oncogene, serine/threonine kinase)**

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival[1,2]. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF[3]. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead[3,4,5].

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)[6,7,8,9,10]. Mutations at V600 belong to class 1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types[4,11]. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions[4]. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and

## Biomarker Descriptions (continued)

N581S/I[4]. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms[12,13,14]. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in exon 11 as well as additional codons flanking V600 in the activation loop[11]. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer[7,10]. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types[15,16,17,18,19]. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation[5,15,17].

Potential relevance: Vemurafenib[20] (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive[4]. BRAF kinase inhibitors including dabrafenib[21] (2013) and encorafenib[22] (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib[22] is approved in combination with cetuximab (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations[4]. Trametinib[23] (2013) and binimetinib[24] (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib[25] (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The PD-L1 antibody, atezolizumab[26], has also been approved in combination with cobimetinib and vemurafenib for BRAF V600 mutation-positive unresectable or metastatic melanoma. The pan-RAF kinase inhibitor DAY-101 was granted breakthrough therapy designation (2020) by the FDA for pediatric patients with advanced low-grade glioma harboring activating RAF alterations[27]. The ERK inhibitor ulixertinib[28] was also granted a fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma[29]. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)[30,31,32,33,34,35,36]. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported[19].

**NRAS (NRAS proto-oncogene, GTPase)**

Background: The NRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes KRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival[37,38,39].

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. NRAS mutations are particularly common in melanomas (up to 25%) and are observed at frequencies of 5-10% in acute myeloid leukemia, colorectal, and thyroid cancers[10,40]. The majority of NRAS mutations consist of point mutations at G12, G13, and Q61[10,41]. Mutations at A59, K117, and A146 have also been observed but are less frequent[7,42].

Potential relevance: Currently, no therapies are approved for NRAS aberrations. The EGFR antagonists, cetuximab[43] and panitumumab[44], are contraindicated for treatment of colorectal cancer patients with NRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)[42]. NRAS mutations are associated with poor prognosis in patients with low-risk myelodysplastic syndrome[45] as well as melanoma[46]. In a phase III clinical trial in patients with advanced NRAS-mutant melanoma, binimetinib improved progression free survival (PFS) relative to dacarbazine with median PFS of 2.8 and 1.5 months, respectively[47].

## Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

### BRAF V600E

| Relevant Therapy | FDA | NCCN | Clinical Trials* |
|---|---|---|---|
| dabrafenib | ◐ | ◐ | ✖ |
| dabrafenib + trametinib | ◐ | ◐ | ✖ |
| binimetinib + encorafenib | ◐ | ● | ✖ |
| cetuximab + encorafenib | ◐ | ○ | ✖ |
| trametinib | ◐ | ✖ | ✖ |
| cobimetinib + vemurafenib | ● | ◐ | ● (II) |
| vemurafenib | ● | ◐ | ● (II) |
| atezolizumab + cobimetinib + vemurafenib | ● | ✖ | ✖ |
| dabrafenib + pembrolizumab + trametinib | ✖ | ● | ✖ |
| encorafenib | ✖ | ● | ✖ |
| encorafenib + panitumumab | ✖ | ○ | ✖ |
| selumetinib | ✖ | ○ | ✖ |
| bempegaldesleukin, nivolumab | ✖ | ✖ | ● (III) |
| dabrafenib, trametinib, ipilimumab, nivolumab | ✖ | ✖ | ● (III) |
| encorafenib, binimetinib, pembrolizumab | ✖ | ✖ | ● (III) |
| sargramostim, nivolumab, ipilimumab | ✖ | ✖ | ● (II/III) |
| abemaciclib + LY3214996 | ✖ | ✖ | ● (II) |
| atezolizumab, cobimetinib, vemurafenib | ✖ | ✖ | ● (II) |
| binimetinib, encorafenib | ✖ | ✖ | ● (II) |
| dabrafenib, nivolumab, trametinib | ✖ | ✖ | ● (II) |
| dabrafenib, trametinib | ✖ | ✖ | ● (II) |
| encorafenib, binimetinib, nivolumab, ipilimumab | ✖ | ✖ | ● (II) |
| ipilimumab, binimetinib, nivolumab, encorafenib | ✖ | ✖ | ● (II) |
| ipilimumab, nivolumab, encorafenib, binimetinib, dabrafenib, trametinib | ✖ | ✖ | ● (II) |
| LXH254 , LTT-462, trametinib, ribociclib | ✖ | ✖ | ● (II) |
| nivolumab, ipilimumab, radiation therapy | ✖ | ✖ | ● (II) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

## Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

### BRAF V600E (continued)

| Relevant Therapy | FDA | NCCN | Clinical Trials* |
|---|---|---|---|
| vemurafenib, cobimetinib | ✖ | ✖ | ● (II) |
| vemurafenib, cobimetinib, atezolizumab | ✖ | ✖ | ● (II) |
| vemurafenib, cobimetinib, ipilimumab, nivolumab | ✖ | ✖ | ● (II) |
| vemurafenib, cobimetinib, surgical intervention, radiation therapy | ✖ | ✖ | ● (II) |
| vemurafenib, dabrafenib | ✖ | ✖ | ● (II) |
| ASTX029 | ✖ | ✖ | ● (I/II) |
| bemcentinib, dabrafenib, pembrolizumab, trametinib | ✖ | ✖ | ● (I/II) |
| dabrafenib, trametinib, antimalarial | ✖ | ✖ | ● (I/II) |
| encorafenib, binimetinib, palbociclib | ✖ | ✖ | ● (I/II) |
| HH-2710 | ✖ | ✖ | ● (I/II) |
| mirdametinib, lifirafenib | ✖ | ✖ | ● (I/II) |
| ABM-1310 | ✖ | ✖ | ● (I) |
| AZD-0364 | ✖ | ✖ | ● (I) |
| BGB-3245 | ✖ | ✖ | ● (I) |
| binimetinib, nivolumab, encorafenib | ✖ | ✖ | ● (I) |
| CART-GD2, vemurafenib | ✖ | ✖ | ● (I) |
| cobimetinib, belvarafenib | ✖ | ✖ | ● (I) |
| cobimetinib, vemurafenib | ✖ | ✖ | ● (I) |
| DAY-101 | ✖ | ✖ | ● (I) |
| DCC-3116, trametinib | ✖ | ✖ | ● (I) |
| E6201 | ✖ | ✖ | ● (I) |
| HL-085, vemurafenib | ✖ | ✖ | ● (I) |
| JSI-1187, dabrafenib | ✖ | ✖ | ● (I) |
| KIN-2787 | ✖ | ✖ | ● (I) |
| PF 07284890, binimetinib | ✖ | ✖ | ● (I) |
| RMC-4630 | ✖ | ✖ | ● (I) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

## Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

### BRAF V600E (continued)

| Relevant Therapy | FDA | NCCN | Clinical Trials* |
|---|---|---|---|
| RO-5126766, everolimus | ✖ | ✖ | ● (I) |
| RX208 | ✖ | ✖ | ● (I) |

### NRAS G13D

| Relevant Therapy | FDA | NCCN | Clinical Trials* |
|---|---|---|---|
| binimetinib | ✖ | ● | ✖ |
| HL-085 | ✖ | ✖ | ● (II) |
| LXH254 , LTT-462, trametinib, ribociclib | ✖ | ✖ | ● (II) |
| ASTX029 | ✖ | ✖ | ● (I/II) |
| HH-2710 | ✖ | ✖ | ● (I/II) |
| mirdametinib, lifirafenib | ✖ | ✖ | ● (I/II) |
| navitoclax, trametinib | ✖ | ✖ | ● (I/II) |
| neratinib, valproic acid | ✖ | ✖ | ● (I/II) |
| trametinib, antimalarial | ✖ | ✖ | ● (I/II) |
| AZD-0364 | ✖ | ✖ | ● (I) |
| BGB-3245 | ✖ | ✖ | ● (I) |
| BI-3011441 | ✖ | ✖ | ● (I) |
| cobimetinib, atezolizumab, belvarafenib | ✖ | ✖ | ● (I) |
| cobimetinib, belvarafenib | ✖ | ✖ | ● (I) |
| DAY-101 | ✖ | ✖ | ● (I) |
| DCC-3116, trametinib | ✖ | ✖ | ● (I) |
| FCN-159 | ✖ | ✖ | ● (I) |
| IN10018, cobimetinib | ✖ | ✖ | ● (I) |
| JSI-1187 | ✖ | ✖ | ● (I) |
| PF-07284892, binimetinib | ✖ | ✖ | ● (I) |
| RMC-4630 | ✖ | ✖ | ● (I) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

## Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

### NRAS G13D (continued)

| Relevant Therapy | FDA | NCCN | Clinical Trials* |
|---|---|---|---|
| RO-5126766, everolimus | ✖ | ✖ | ● (I) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

## Clinical Trials Summary

### BRAF V600E

| NCT ID | Title | Phase |
|---|---|---|
| NCT02224781 | DREAMseq (Doublet, Randomized Evaluation in Advanced Melanoma Sequencing) a Phase III Trial | III |
| NCT03898908 | Phase II, Multicentre Clinical Trial to Evaluate the Activity of Encorafenib and Binimetinib Administered Before Local Treatment in Patients With BRAF Mutant Melanoma Metastatic to the Brain. | II |
| NCT02231775 | Neoadjuvant and Adjuvant Dabrafenib and Trametinib in Patients With Clinical Stage III Melanoma (Combi-Neo) | II |
| NCT04722575 | "NEOadjuvant Plus Adjuvant Therapy With Combination or Sequence of Vemurafenib, cobImetinib, and atezolizuMab in Patients With High-risk, Surgically Resectable BRAF Mutated and Wild-type Melanoma" | II |
| NCT02968303 | Phase II Study With COmbination of Vemurafenib With Cobimetinib in B-RAF V600E/K Mutated Melanoma Patients to Normalize LDH and Optimize Nivolumab and Ipilimumab therapy | II |
| NCT04655157 | A Multi-Center Phase I/II Open Label Study to Evaluate Safety and Efficacy in Participants With Metastatic BRAF-mutant Melanoma Treated With Encorafenib With and Without Binimetinib in Combination With Nivolumab and Low-dose Ipilimumab. (QUAD 01: Quadruple Therapy in Melanoma) | I/II |
| NCT04190628 | A Phase I, First-In-Human, Multicenter, Open-Label Study of ABM-1310, Administered Orally in Adult Patients With Advanced Solid Tumors | I |
| No NCT ID | Phase I Study of Safety and Immune Effects of an Escalating Dose of Autologous GD2 Chimeric Antigen Receptor-Expressing Peripheral Blood T cells in Patients with Metastatic Melanoma | I |
| NCT03554083 | Neoadjuvant Therapy for Patients With High Risk Stage III Melanoma: A Pilot Clinical Trial | II |
| NCT04511013 | A Randomized Phase II Trial of Encorafenib + Binimetinib + Nivolumab vs Ipilimumab + Nivolumab in BRAF-V600 Mutant Melanoma With Brain Metastases | II |
| NCT03514901 | An Evaluation of the Efficacy Beyond Progression of Vemurafenib Combined With Cobimetinib Associated With Local Treatment Compared to Second-line Treatment in Patients With BRAFV600 Mutation-positive Metastatic Melanoma in Focal Progression With First-line Combined Vemurafenib and Cobimetinib. | II |
| NCT04720768 | A Phase IB Study to Evaluate the Safety, Tolerability and Preliminary Efficacy of the Combination of Encorafenib, Binimetinib and Palbociclib in Patients With BRAF-mutant Metastatic Melanoma (The CELEBRATE Study) | I/II |

## Clinical Trials Summary (continued)

### BRAF V600E (continued)

| NCT ID | Title | Phase |
|---|---|---|
| NCT04249843 | A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumors | I |
| NCT04741997 | A Randomized Pilot Trial of Adjuvant Therapy Based on Pathologic Response After Neoadjuvant Encorafenib and Binimetinib in Advanced Melanoma | I |
| NCT03284502 | A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors | I |
| NCT03543969 | Pilot Study of Adaptive BRAF-MEK Inhibitor Therapy for Advanced BRAF Mutant Melanoma | I |
| NCT04418167 | A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations | I |
| NCT04543188 | A Two-Part, Phase Ia/b, Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF 07284890 (ARRY 461) in Participants with Braf V600 Mutant Solid Tumors with and without Brain Involvement | I |
| NCT03635983 | A Phase III, Randomized, Open-label Study of NKTR-214 Combined With Nivolumab Versus Nivolumab in Participants With Previously Untreated Unresectable or Metastatic Melanoma | III |
| NCT02339571 | Randomized Phase II/III Study of Nivolumab Plus Ipilimumab Plus Sargramostim Versus Nivolumab Plus Ipilimumab in Patients With Unresectable Stage III or Stage IV Melanoma | II/III |
| NCT02910700 | A Phase II Study of the TRIplet Combination of Dabrafenib, Nivolumab, and Trametinib in Patients With Metastatic Melanoma (TRIDeNT) | II |
| NCT03563729 | Efficacy Of Immunotherapy In Melanoma Patients With Brain Metastases Treated With Steroids | II |
| NCT04892017 | A Phase I, First-in-Human Study of DCC-3116 as a Single Agent and in Combination With Trametinib in Patients With Advanced or Metastatic Solid Tumors With RAS or RAF Mutations. | I |
| NCT03332589 | A Phase I Study Of E6201 For The Treatment Of Central Nervous System Metastases (CNS) From BRAF Or MEK-Mutated Metastatic Melanoma | I |
| NCT03905148 | A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors | I/II |
| NCT04657991 | A Phase III, Randomized, Double-blind Study of Encorafenib and Binimetinib Plus Pembrolizumab Versus Placebo Plus Pembrolizumab in Participants With BRAF V600E/K Mutation-Positive Metastatic OR Unresectable Locally Advanced Melanoma | III |
| NCT02693535 | Targeted Agent and Profiling Utilization Registry (TAPUR) Study. | II |
| NCT03235245 | Combination of Targeted Therapy (Encorafenib and Binimetinib) Followed by Combination of Immunotherapy (Ipilimumab and Nivolumab) vs Immediate Combination of Immunotherapy in Patients With Unresectable or Metastatic Melanoma With BRAF V600 Mutation : an EORTC Randomized Phase II Study (EBIN) | II |
| NCT04591431 | The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy | II |

## Clinical Trials Summary (continued)

### BRAF V600E (continued)

| NCT ID | Title | Phase |
| --- | --- | --- |
| No NCT ID | Phase I study of the safety, tolerability, pharmacokinetics and preliminary efficacy of RX208 in patients with advanced malignant solid tumors | I |
| NCT03297606 | Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial | II |
| NCT04417621 | A Randomized, Open-label, Multi-arm, Two-part, Phase II Study to Assess Efficacy and Safety of Multiple LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic BRAFV600 or NRAS Mutant Melanoma | II |
| NCT03155620 | NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol | II |
| NCT03220035 | NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of Vemurafenib in Patients With Tumors Harboring Braf V600 Mutations | II |
| NCT03754179 | A lead-in Phase I Followed by a Phase II Clinical Trial on the Combination of Dabrafenib, Trametinib and the Autophagy Inhibitor Hydroxychloroquine in BRAF/MEK Inhibitor-pretreated Patients With Advanced BRAF V600 Mutant Melanoma | I/II |
| NCT03781219 | A Phase I, Single Arm, Dose Escalation Study to Evaluate Safety, Pharmacokinetics and Preliminary Efficacy of HL-085 Plus Vemurafenib in Patients With BRAF V600 Mutant Advanced Solid Tumor | I |
| NCT04534283 | A Phase II Basket Trial of an ERK1/2 Inhibitor (LY3214996) in Combination With Abemaciclib for Patients Whose Tumors Harbor Pathogenic Alterations in BRAF, RAF1, MEK1/2, ERK1/2, and NF1 | II |
| NCT04423185 | Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China | II |
| NCT03239015 | Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event | II |
| NCT02872259 | A Phase Ib/II Randomised Open Label Study of BGB324 in Combination With Pembrolizumab or Dabrafenib/Trametinib Compared to Pembrolizumab or Dabrafenib/Trametinib Alone, in Patients With Advanced Non-resectable (Stage IIIc) or Metastatic (Stage IV) Melanoma. | I/II |
| NCT02407509 | A Phase I Trial of RO5126766 (a Dual RAF/MEK Inhibitor) Exploring Intermittent, Oral Dosing Regimens in Patients With Solid Tumours or Multiple Myeloma, With an Expansion to Explore Intermittent Dosing in Combination With Everolimus | I |
| NCT03340129 | A Phase II, Open Label, Single Arm Study of Ipilimumab and Nivolumab With Salvage Radiotherapy in Patients With Melanoma Brain Metastases | II |
| NCT04913285 | A Phase I/Ib Open-Label, Multicenter, Two Part Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Antitumor Activity of KIN-2787 in Subjects With BRAF Mutation Positive Solid Tumors | I |
| NCT04198818 | A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors | I/II |
| NCT03429803 | A Phase I Study of TAK-580 (MLN2480) for Children With Low-Grade Gliomas and Other RAS/RAF/MEK/ERK Pathway Activated Tumors | I |
| NCT03634982 | A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors | I |
| NCT03520075 | A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors | I/II |

# Clinical Trials Summary (continued)

## BRAF V600E (continued)

| NCT ID | Title | Phase |
|---|---|---|
| NCT04305249 | A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies | I |

## NRAS G13D

| NCT ID | Title | Phase |
|---|---|---|
| NCT03979651 | MEK and Autophagy Inhibition in Metastatic/Locally Advanced, Unresectable Neuroblastoma RAS (NRAS) Melanoma: A Phase Ib/II Trial of Trametinib Plus Hydroxychloroquine in Patients With NRAS Melanoma | I/II |
| No NCT ID | A single-arm, multi-center phase II clinical study to evaluate the efficacy and safety of HL-085 capsules in the treatment of patients with advanced melanoma with NRAS mutations | II |
| NCT03284502 | A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors | I |
| NCT04892017 | A Phase I, First-in-Human Study of DCC-3116 as a Single Agent and in Combination With Trametinib in Patients With Advanced or Metastatic Solid Tumors With RAS or RAF Mutations. | I |
| NCT03932253 | A Phase Ia/Ib Clinical Study to Evaluate the Safety, Pharmacokinetics (PK) and Preliminary Anti-tumor Activity of FCN-159 in Patients With Advanced Melanoma Harboring NRAS-aberrant (Ia) and NRAS-mutant (Ib) | I |
| NCT04109456 | A Phase Ib, Open-label Clinical Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Antitumor Activities of IN10018 as Monotherapy and Combination Therapy in Subjects With Metastatic Melanoma | I |
| NCT03905148 | A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors | I/II |
| NCT02079740 | An Open Label, Two-Part, Phase Ib/II Study to Investigate the Safety, Pharmacokinetics, Pharmacodynamics, and Clinical Activity of the MEK Inhibitor Trametinib and the BCL2-Family Inhibitor Navitoclax (ABT-263) in Combination in Subjects With KRAS or NRAS Mutation-Positive Advanced Solid Tumors | I/II |
| NCT04742556 | A Phase I Open-label Trial of BI 3011441 in Japanese Patients With NRAS/KRAS Mutation Positive Advanced, Unresectable or Metastatic Refractory Solid Tumours | I |
| NCT04417621 | A Randomized, Open-label, Multi-arm, Two-part, Phase II Study to Assess Efficacy and Safety of Multiple LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic BRAFV600 or NRAS Mutant Melanoma | II |
| NCT03919292 | Phase 1/2 Study of Neratinib and Divalproex Sodium (Valproate) in Advanced Solid Tumors, With an Expansion Cohort in Ras-Mutated Cancers | I/II |
| NCT04249843 | A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumors | I |

## Clinical Trials Summary (continued)

### NRAS G13D (continued)

| NCT ID | Title | Phase |
|---|---|---|
| NCT04835805 | A Phase Ib, Open-Label, Multicenter Study to Evaluate the Safety, Pharmacokinetics, and Activity of Belvarafenib as a Single Agent and in Combination With Either Cobimetinib or Cobimetinib Plus Atezolizumab in Patients With NRAS-Mutant Advanced Melanoma Who Have Received Anti-PD-1/PD-L1 Therapy | I |
| NCT04418167 | A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations | I |
| NCT02974725 | A Phase Ib, Open-label, Multicenter Study of Oral LXH254-centric Combinations in Adult Patients With Advanced or Metastatic KRAS or BRAF Mutant Non-Small Cell Lung Cancer or NRAS Mutant Melanoma | I |
| NCT02407509 | A Phase I Trial of RO5126766 (a Dual RAF/MEK Inhibitor) Exploring Intermittent, Oral Dosing Regimens in Patients With Solid Tumours or Multiple Myeloma, With an Expansion to Explore Intermittent Dosing in Combination With Everolimus | I |
| NCT04800822 | A Phase I Open-Label, Multi-Center, Dose Escalation and Dose Expansion Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of PF-07284892 (ARRY-558) as a Single Agent and in Combination Therapy in Participants With Advanced Solid Tumors | I |
| NCT04198818 | A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors | I/II |
| NCT03429803 | A Phase I Study of TAK-580 (MLN2480) for Children With Low-Grade Gliomas and Other RAS/RAF/MEK/ERK Pathway Activated Tumors | I |
| NCT03634982 | A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors | I |
| NCT03520075 | A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors | I/II |
| NCT04305249 | A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies | I |

## Assay Information

**Methodology, Scope, and Application**:The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

**Analytical Sensitivity and Specificity**:The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

**Hereditary and Germline Mutations**:This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

**Report Content**:Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

**Genes assayed for hotspot mutations**:
*AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1*

**Genes assayed with full exon coverage**:
*ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2*

**Genes assayed for copy number variations**:
*AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT*

**Genes assayed for detection of fusions**:
*AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT*

## DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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